Recall. $\mathcal{O}$ is noetherian, artinian i.e. all objects have finite length.

Thm $\mathcal{C}$ is noeth, artinian, every M has finite composition series

$$M_0 = 0 \subseteq M_1 \subseteq \dots \subseteq M_n = M$$

M_i / M_{i-1} simple

These simples + multiplicities independent of filtration chosen.

Sketch: induct on the minimal length n of filtration of M.

$n=1$: simple, obvious. ($\Rightarrow$ all series same length)

induct: $M_0 \subseteq M \dots \subseteq M_n$ minimal

$N_0 \subseteq \dots \subseteq N_r$

i.e. M/M_0

$N_0 \subseteq \dots N_{i-1} \subseteq N_{i+1}/M_0 \subseteq \dots \subseteq M/M_0$

exercise

① If $N_0 = M_0$ done. kill both.

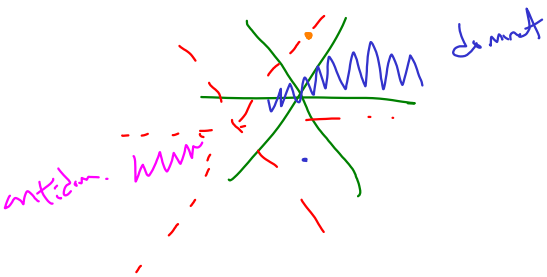
② If not, eventually $M_0 \subseteq N_i$ (smallest i)

$\Rightarrow N_r / N_{i-1} = M_0$ both simple

$\searrow$ (ii), induct.

Upshot. Think about M as 'built' from simples

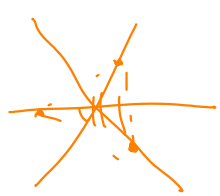
Fact.
- Simples in $\mathcal{O}$ L_λ
- Standards/Vermas $M(\lambda) = \Delta_\lambda$ projective if $\lambda+\rho$ dominant (positive) irreducible if $\lambda+\rho$ antidominant (neg.)



$\chi : \mathfrak{Z}_{\mathfrak{g}} \to \mathbb{C}$ central character

Def. $\mathcal{O}_\chi \subseteq \mathcal{O}$ where $\mathfrak{Z}_{\mathfrak{g}} - \chi$ acts locally nilpotently.

i.e. $z \in \mathfrak{Z}_{\mathfrak{g}}$, $(z - \chi(z))^n_m = 0$ for large n, for each m.



distinguish between being on shifted walls vs not

- χ ~~is dominant if $\chi = [\lambda]$ for $\lambda+\rho$ dominant~~

$\lambda^{+\rho}$ regular if $\lambda^{+\rho}$ is not on any reflection hyperplane.

Results.

$$\mathcal{O} = \bigoplus_{\substack{\chi = [\lambda] \\ \lambda \in \mathfrak{h}^*}} \mathcal{O}_\chi$$

Furthermore, if χ is regular integral, then $\mathcal{O}_\chi$ indecomposable.

Prop: Hard to cut locally finite exam

$\Rightarrow M = \bigoplus_{x \in \mathcal{Z}(\mathfrak{g})} M_x$

Counterexample: $A = k[x]$, $\mathcal{C} = A$-mod.

$M = k[x] \supseteq x k[x] \supseteq x^2 k[x] \supseteq \dots$

simples $\boxed{}$ $\mathbb{A}^1$

Second claim:

- if λ is dom. integral. $\Theta_\lambda \Delta_\lambda$ is indecomposable (gen by vector)
- every $L_{w\cdot\lambda}$ Δ_λ a subquotient

Recall: basic problem: find $a_{\lambda\mu}$ $b_{\lambda\mu}$ such that

$$[\Delta_\lambda] = \sum_\mu a_{\lambda\mu}[L_\mu] \qquad [L_\lambda] = \sum_\mu b_{\lambda\mu}[\Delta_\mu]$$

$$= \sum_{w \in W} a_{\lambda, w\cdot\lambda}[L_{w\cdot\lambda}]$$

$a_{\lambda,w}$

- all $=0$ if λ, μ not W-linked ($\Rightarrow$ forms on $\mathcal{O}_\chi$)
- $=0$ if $\lambda \leq \mu$ ($\Rightarrow$ "upper triangular")
- $\exists$ $\mathcal{O}_\lambda$, $\mathcal{O}_{\lambda'}$ same constants for λ regular integral.

$a_{\lambda, w\cdot\lambda} = a_w$ if λ integral regular.

Claim: $a_w = (-1)^{\ell(w)}$

Proof: Use symmetry + $ch(L_\lambda)$ W-invariant (non-linked antis)

weight space dim

(finite-dim)

$$ch(L_\lambda) = \sum a_y \, ch(\Delta_{y\cdot\lambda}) = \sum a_y \frac{e^{y(\lambda+\rho)-\rho}}{\prod_{\alpha \in \Delta^+} (1 - e^{-\alpha})}$$

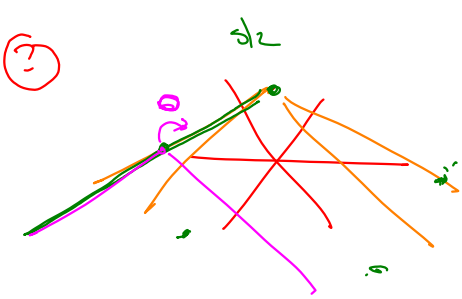
highest weight

neg. simple roots.

Act by s_α, show get (-1).

Claim: every irred. appears... ?

Δ_λ has every highest weight...



Def: $\tau : \mathfrak{g} \to \mathfrak{g}$ Cartan involution $\quad \tau = -1$ on $\mathfrak{h}$ $\quad [e,f] = h$

swaps $\mathfrak{n}^+, \mathfrak{n}^-$

$e_\alpha \mapsto f_\alpha$

$M \in \mathcal{O}$ the contragredient dual

graded dual of M is $\ker$

k-lin dual.

Write: $M = \bigoplus_{\lambda \in \mathfrak{h}^*} M^\lambda \rightsquigarrow M^\vee = \bigoplus_{\lambda \in \mathfrak{h}^*} (M^\lambda)^*$

$\mathfrak{g}$ acts by antipode twisted by τ.

$x \in \mathfrak{g}, \quad (x \cdot f)(m) = f(-\tau(m))$

Why $-$ upper?

$[x,y] = xy - yx$

$(xy - yx) \cdot f(m)$

$= f(yxm - xym) = -$

order swapped

$U\mathfrak{g}$ Hopf algebra.

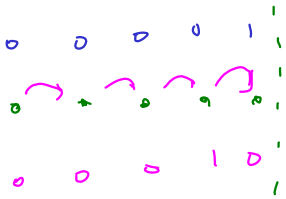
$\omega : x \mapsto 1 \otimes x + x \otimes 1 \quad x$ primitive

antipode: $x \mapsto -x$.

Why twist by τ? Exercise

So get highest weight mod, not lowest weight module.

$\rightsquigarrow$ in $\mathcal{O}$



Exercise: $(M^\vee)^\vee = M$, so $(-)^\vee : \mathcal{O} \xrightarrow{\sim} \mathcal{O}^{op}$

$L_\lambda^\vee = L_\lambda$

Def: Dual Verma: costandard

$\nabla_\lambda = \mathrm{Hom}_{U\mathfrak{b}}(U\mathfrak{g}, \mathbb{C}_\lambda)$

- $U\mathfrak{b}$ acts on right
- $U\mathfrak{g}$ acts through antipode τ-twist.

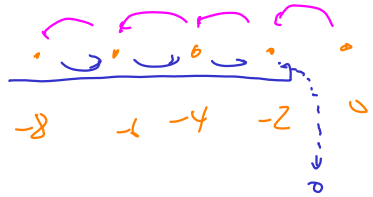
Exercise: Show that

$\nabla_\lambda \cong \Delta_\lambda^\vee \implies \Delta_\lambda^\vee$ has maximal proper unique irred. submodule

(composition series reversed)

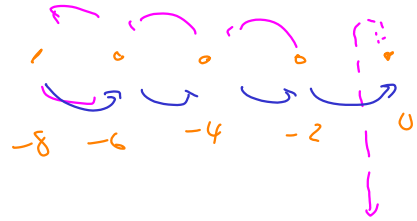
Ex: sl_2

Δ_λ



L_∞ quotient

∇_λ



L_0 sub.

Lemma: $\mathrm{Inf}_{\mathfrak{h}}^{\mathfrak{b}} : \mathrm{Rep}(\mathfrak{g}) \to \mathrm{Rep}(\mathfrak{b})$ $\qquad \mathfrak{b} \to \mathfrak{b}/[\mathfrak{b},\mathfrak{b}] \cong \mathfrak{h}$

$[\mathfrak{b},\mathfrak{b}] = \mathfrak{n}$ = kernel

adjunctions: $((-)_{\mathfrak{n}}, \mathrm{Inf})$

$(\mathrm{Inf}, (-)^{\mathfrak{n}})$

$\mathrm{Res}_{\mathfrak{b}}^{\mathfrak{g}} : \mathrm{Rep}(\mathfrak{g}) \to \mathrm{Rep}(\mathfrak{b})$

?

$(\mathrm{cInd}, \mathrm{Res})$ $\qquad \mathrm{cInd}\, V = U\mathfrak{g} \otimes_{U\mathfrak{b}} V$

adjunction

$(\mathrm{Res}, \mathrm{Ind})$ $\qquad \mathrm{Ind}\, V = \mathrm{Hom}_{U\mathfrak{b}}(U\mathfrak{g}, V)$ $\qquad$ twist τ

$$\mathrm{Hom}_{\mathfrak{g}}(\Delta_\lambda, M) = \mathrm{Hom}_{\mathfrak{g}}(\mathrm{cInd}\,\mathrm{Inf}\, \mathbb{C}_\lambda, M)$$

$$= \mathrm{Hom}_{\mathfrak{b}}(\mathrm{Inf}\, \mathbb{C}_\lambda, \mathrm{Res}\, M)$$

$$= \mathrm{Hom}_{\mathfrak{h}}(\mathbb{C}_\lambda, (\mathrm{Res}\, M)^{\mathfrak{n}})$$

highest weight vectors of M weight λ.

$$\mathrm{Hom}(N, \nabla_\lambda) = \mathrm{Hom}_{\mathfrak{h}}((\mathrm{Res}\, M)_{\mathfrak{n}^-}, \mathbb{C}_\lambda)$$

??